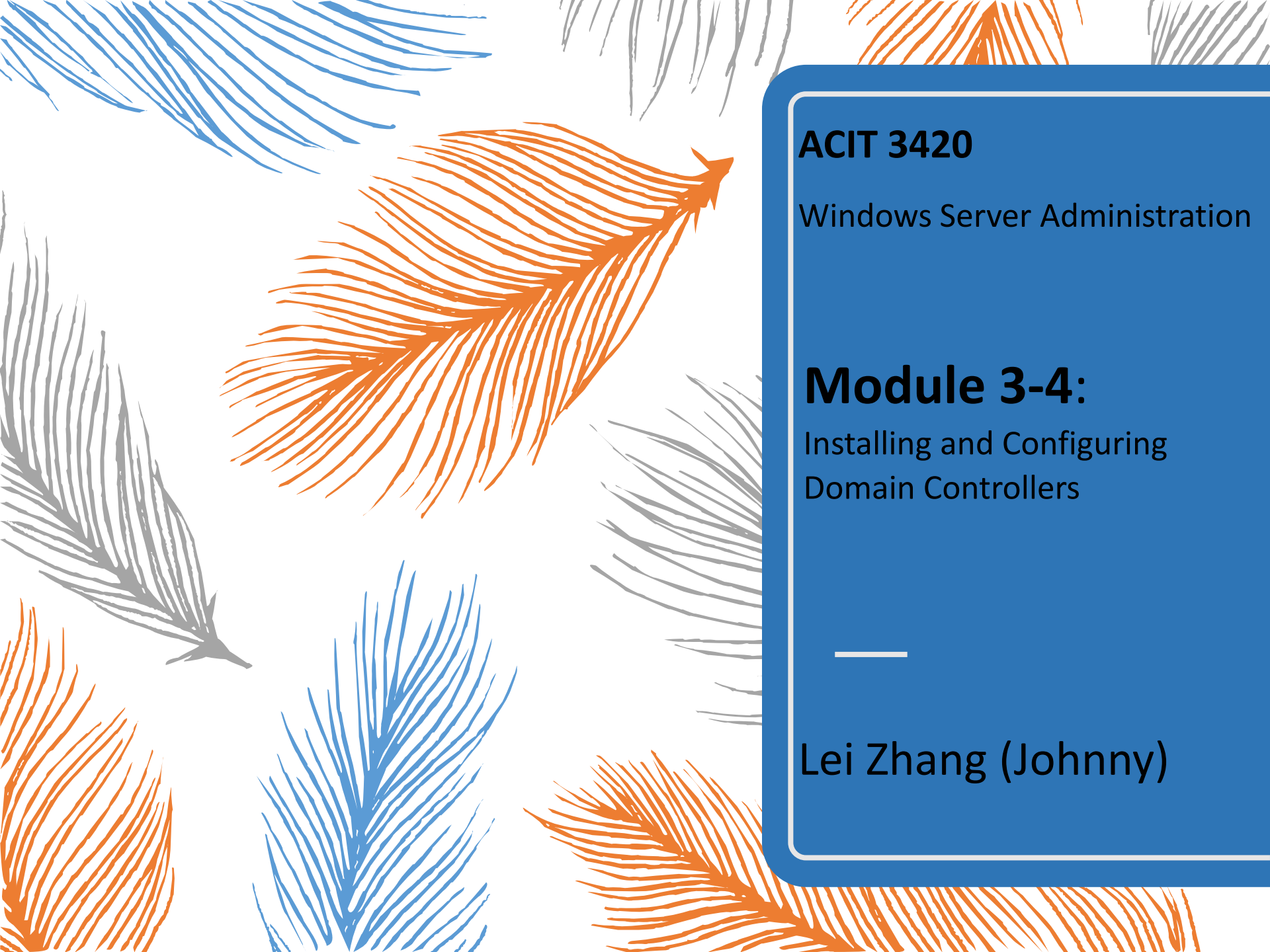


ACIT 3420

Windows Server Administration

Module 3-4:

Installing and Configuring Domain Controllers

Lei Zhang (Johnny)

Learning Objectives

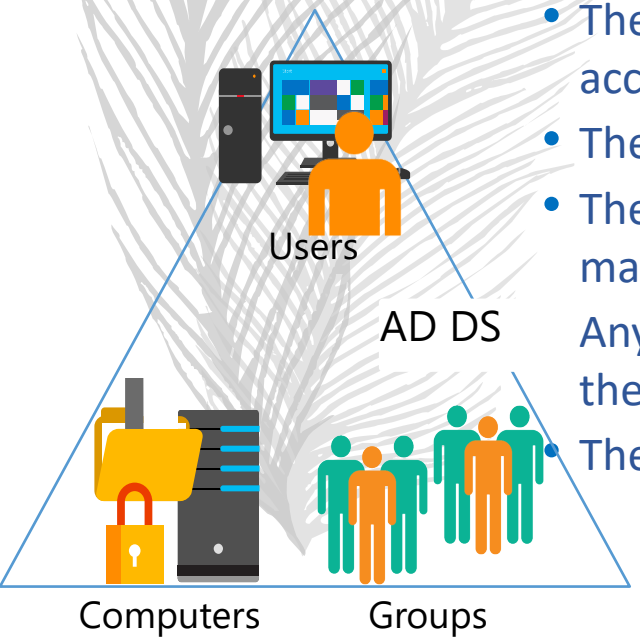
- Terminologies (Partition, scheme, tree, forest, GC, site, OU, domain, subdomain…)
- Deploying Active Directory Domain Services (AD DS)
- Join Domain (GUI)
- Troubleshooting

Understanding Active Directory

Active Directory Domain Services **(AD DS)** is a specialized type of database called a directory. It does the following:

- Stores information about the identities of users, computers, and services
- Provides authentication for users and computers
- Provides authorization for users and computers to access network resources

What is an AD DS?



- AD DS requires one or more domain controllers
- All domain controllers hold a copy of the domain database, which is continually synchronized (Ex. Exchange Problem)
- The domain is the context within which user accounts, computer accounts, and groups are created
- The domain is a replication boundary
- The domain is an administrative center for configuring and managing objects (Ex. Change password)

  Any domain controller can authenticate any sign-in anywhere in the domain (Ex. Logon BBY and DTC)
- The domain provides authorization

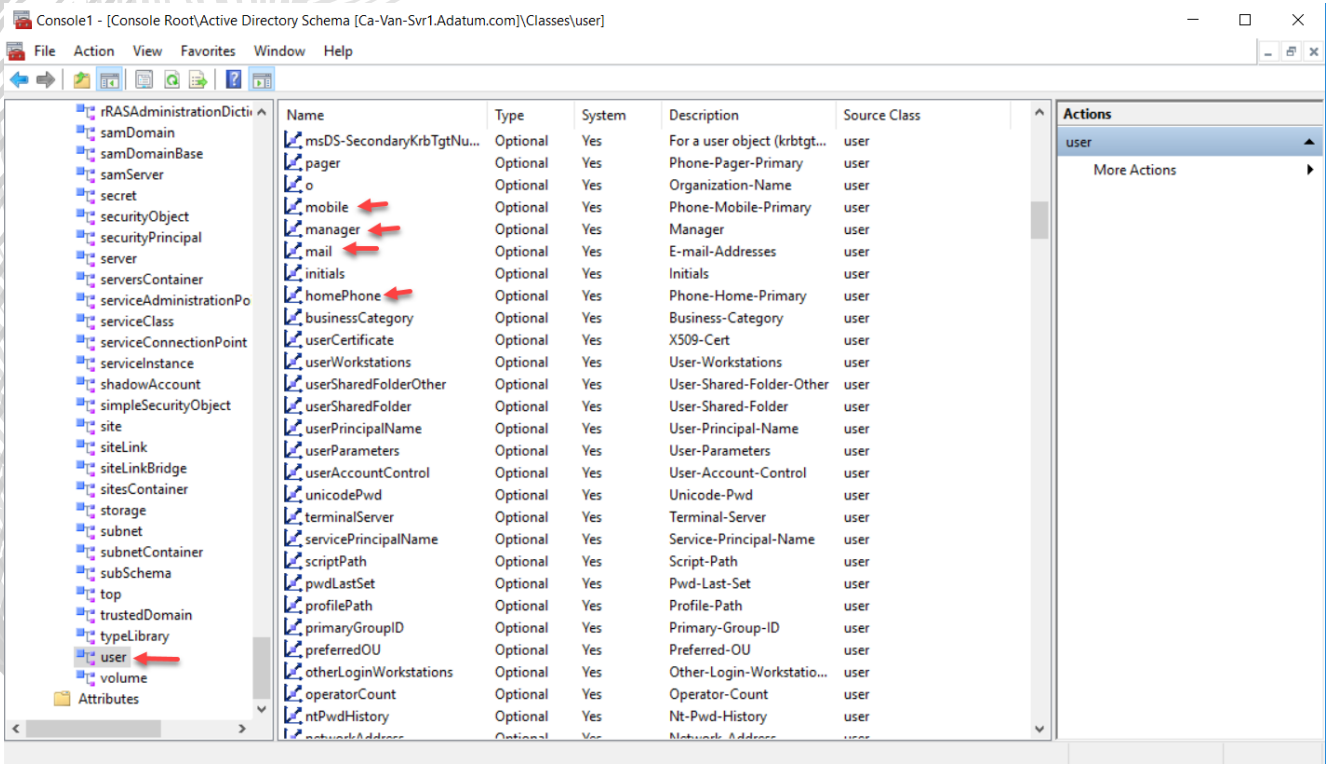
Active Directory Basics

- Domain controllers (DCs)
  - Servers that have the AD DS server role installed
  - Contain writable copies of information in Active Directory
- Member servers
  - Servers on a network managed by Active Directory that do not have Active Directory installed
- Domain
  - Container that holds information about all network resources that are grouped within it
  - Every resource is called an object

Active Directory Basics (cont.)

Schema

- The AD DS schema is the component that defines all the object classes and attributes that AD DS uses to store data.
- Enable Active Directory Schema snap-in
  - ***Run: regsvr32 schmmgmt.dll -> Run: mmc ->File->add/remote Snape-in->add Active Directory Schema***



Active Directory Basics (cont.)

- Partition
  - A partition is a portion of the AD DS database.
  - Although the database is one file named ntds.dit, different partitions contain different data.
    - *Schema partition,*
    - *configuration partition,*
    - *domain partition*
    - *application partition*
  - AD DS database (%systemroot%\NTDS)
    - *ntds.dit*
      - Contains all AD DS partitions and objects
      - Main AD DS database file
    - *EDB*.log : Transaction logs*
    - *EDB.chk : Database checkpoint file*
    - *Edbres00001.jrs (10 MB) & Edbres00002 (10 MB)*
      - Reserve transaction log file that allows the directory to process transactions if the server runs out of disk space

The Active Directory Architecture

- Active Directory is a hierarchical directory service, based on the domain, which is scalable in both directions.
- You can subdivide a domain into organizational units and populate it with objects.
- You can create multiple domains and group them into sites, trees, and forests.
- AD DS provides a highly flexible architecture that can accommodate the smallest and the largest organizations.

Objects and Attributes

- Consists of **objects**, each of which represents a logical or physical resource.
- Each object consists of **attributes** which store information about the object.
- Different objects have different attributes, depending on their function (Ex. Modify Schema for Applications).

Classes of Objects

A **container object** can have other objects subordinate to it:

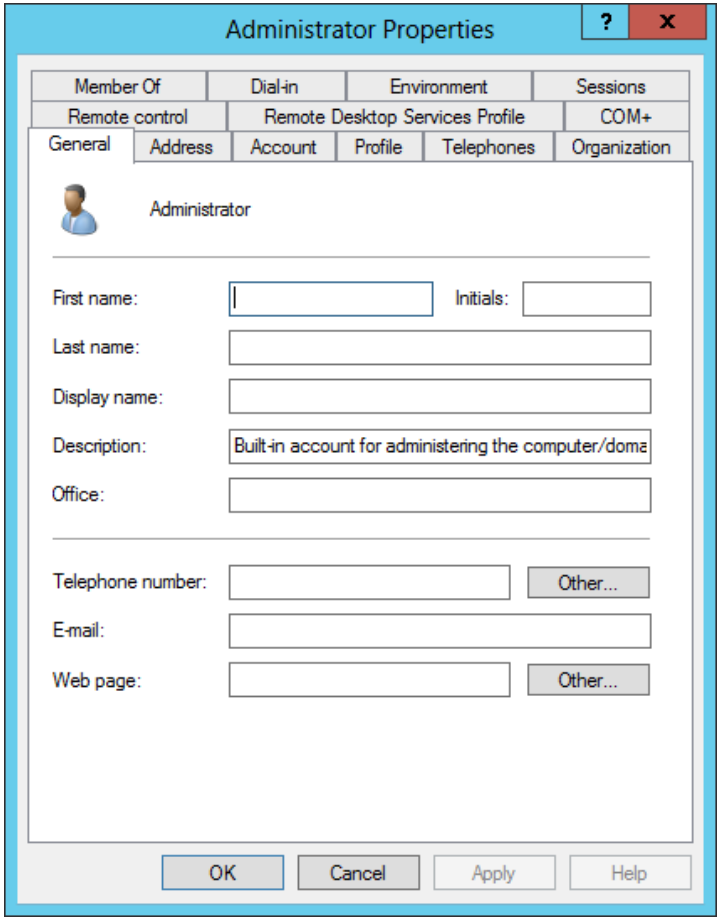
- Domain
- Organizational unit

A **leaf object** cannot have subordinate objects:

- Users
- Computers
- Groups
- Applications
- Network resources

Objects and Attributes

The attributes of a user object, as displayed in its Properties sheet

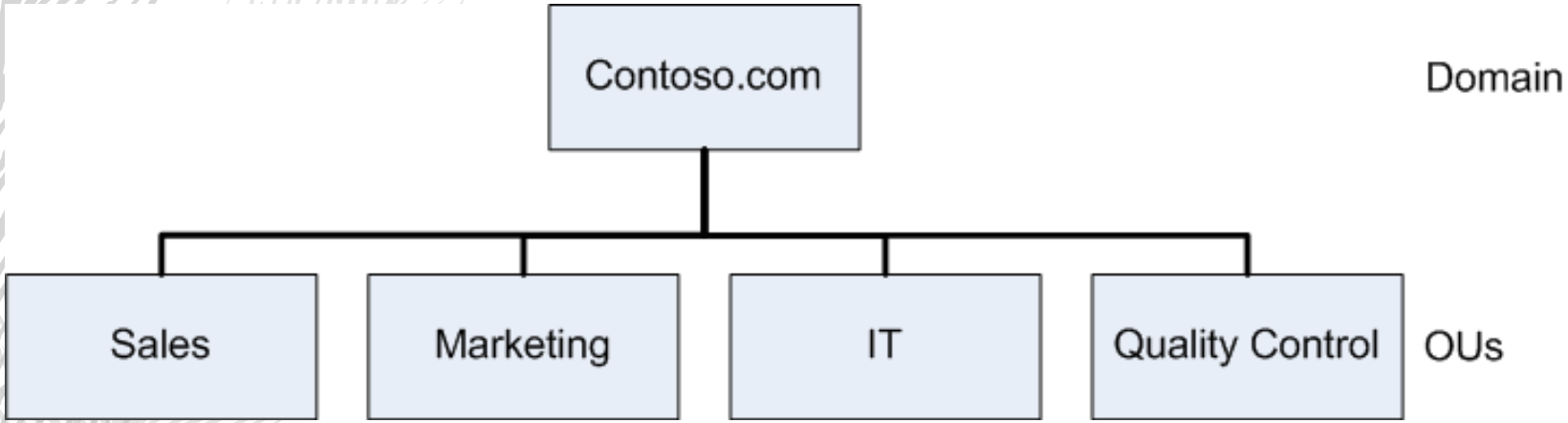


Understanding Active Directory: Logical Components

The *logical* components (which administrators create, organize, and manage) include:

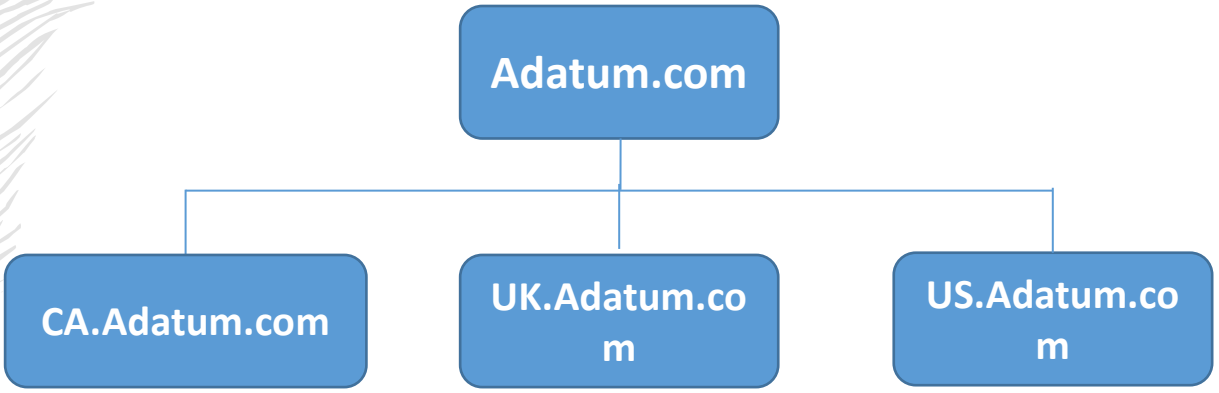
- ***Organizational units (OUs)***: Containers in a domain that allow you to organize and group resources for easier administration, including providing and delegating administrative rights.
- ***Domains***: Administrative boundaries for users and computers that are stored in a common directory database.
- ***Domain trees***: Collections of domains that are grouped together in hierarchical structures and that share a common root domain.
- ***Forests***: Collections of domain trees that share a common AD DS.

Organizational Units



Domain Tree

- When you create your first domain on an Active Directory network, you are creating the root of a **domain tree**.
- Contains one or more domains that are in a common relationship
- Tree has the following characteristics:
  - Two-way trust relationships exist between parent domains and child domains
  - All domains in a single tree use the same schema for all types of common objects
  - All domains use the same global catalog



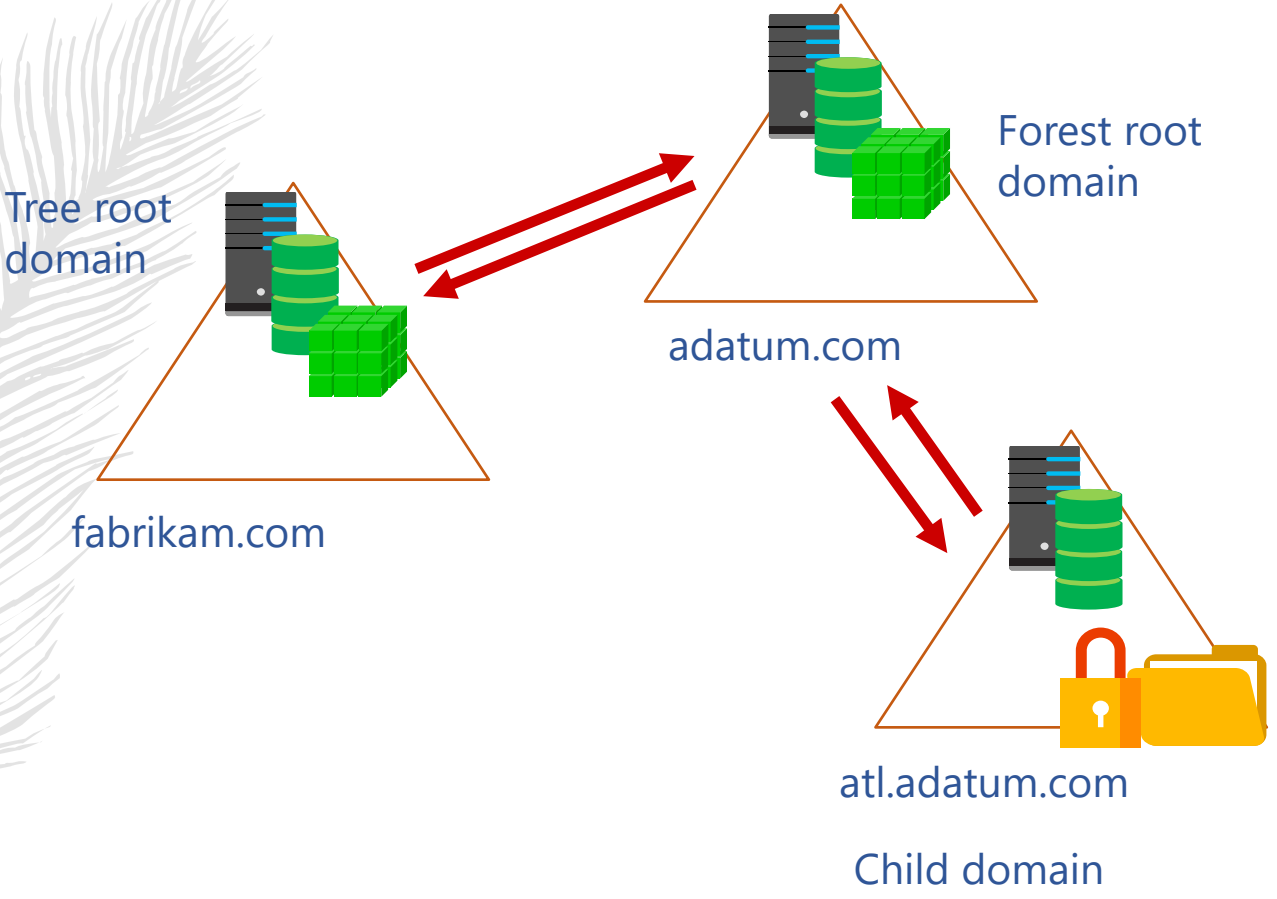
Domain Trees (cont.)

- You can add as many domains to the tree as you need.
- Each domain in a tree is a separate security entity with its own separate Group Policy settings, permissions, and user accounts.
- Unlike OUs, subdomains in a tree do not inherit permissions and policies from their parent domains.
- Domains in the same tree have two-way trusts between parent domains and child domains, which means that an administrator of a particular domain can grant any user in the tree access to that domain's resources.

Forests

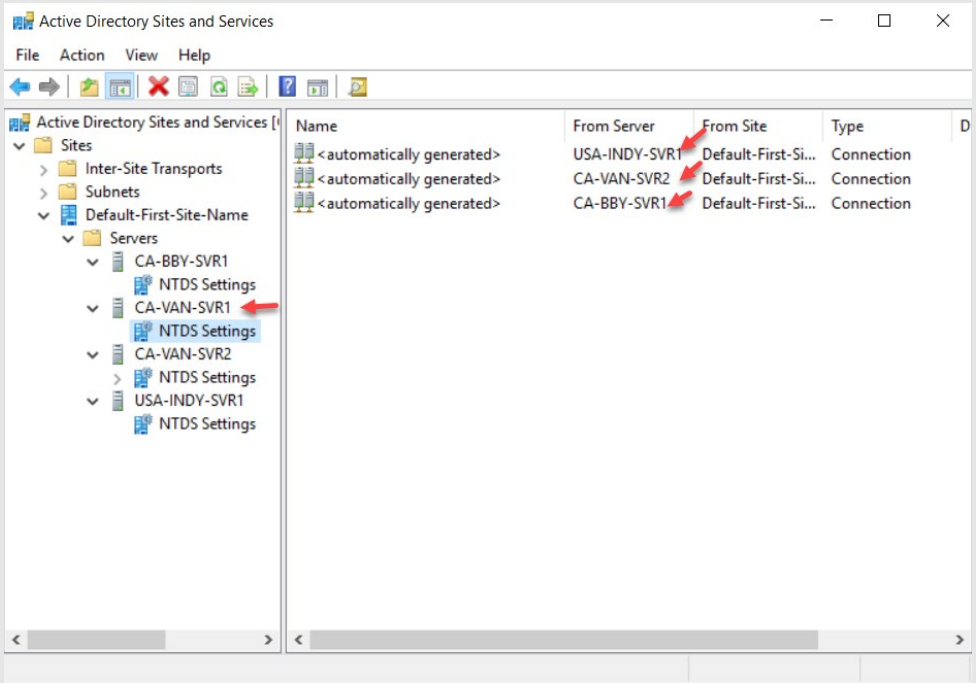
- An Active Directory **forest** consists of one or more separate domain trees, which have the same two-way trust relationships between them as two domains in the same tree.
- When you create the first domain on an Active Directory network, you are creating a new forest, and that first domain becomes the **forest root domain**.

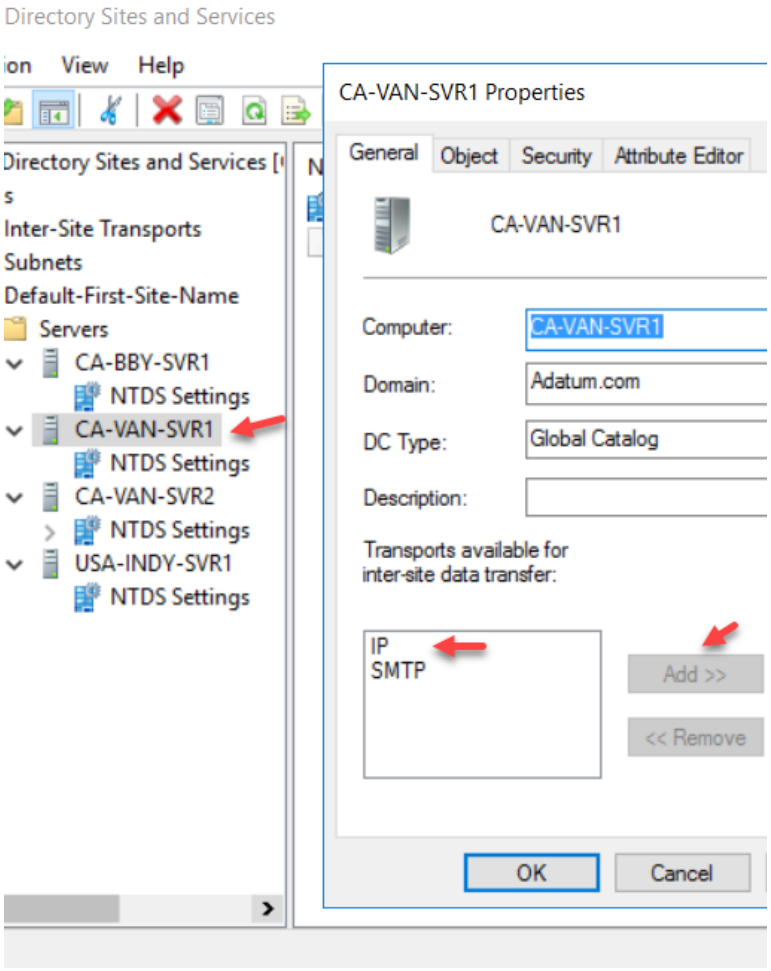
AD DS forest


Tree root domain
Forest root domain
adatum.com
fabrikam.com
atl.adatum.com
Child domain

Site

- A TCP/IP-based concept (container) within Active Directory that is linked to IP subnets
- A site has the following functions:
  - *Reflects one or more interconnected subnets*
  - *Reflects the physical aspect of the network*
  - *Is used for DC replication*
  - *Is used to enable a client to access the DC that is physically closest*
  - *Is composed of only two types of objects, servers and configuration objects*
- Sites are based on connectivity and replication functions

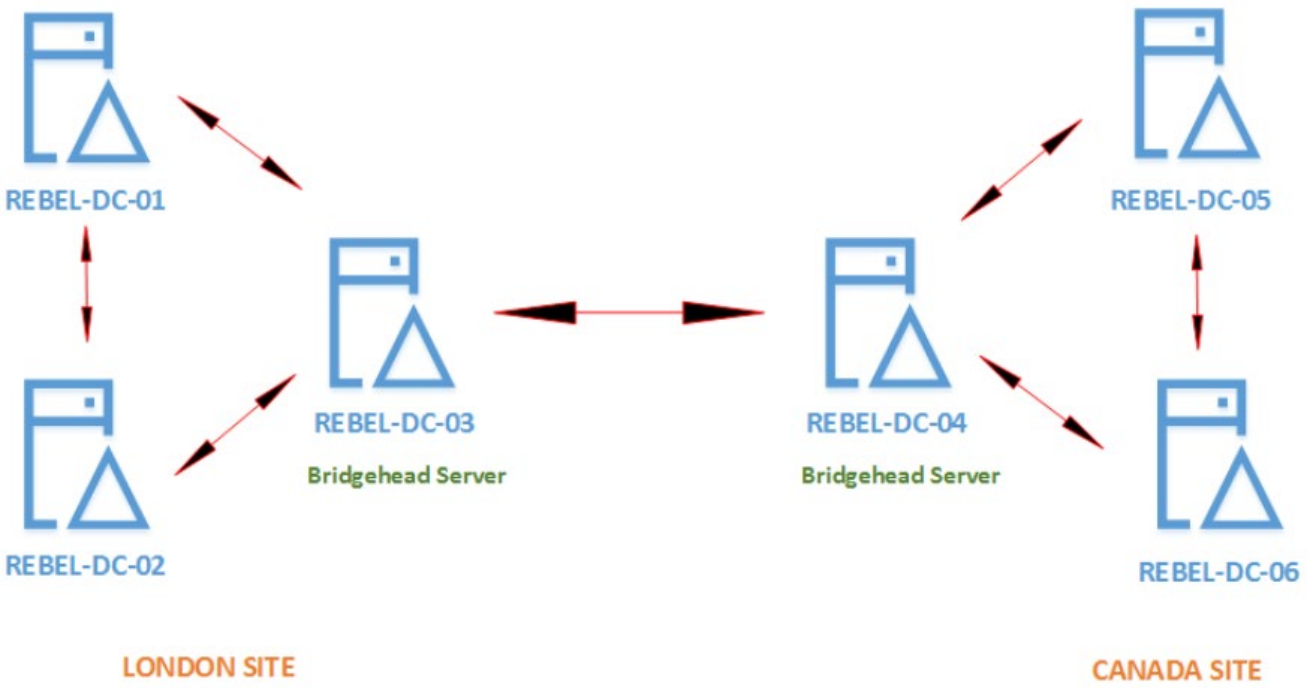




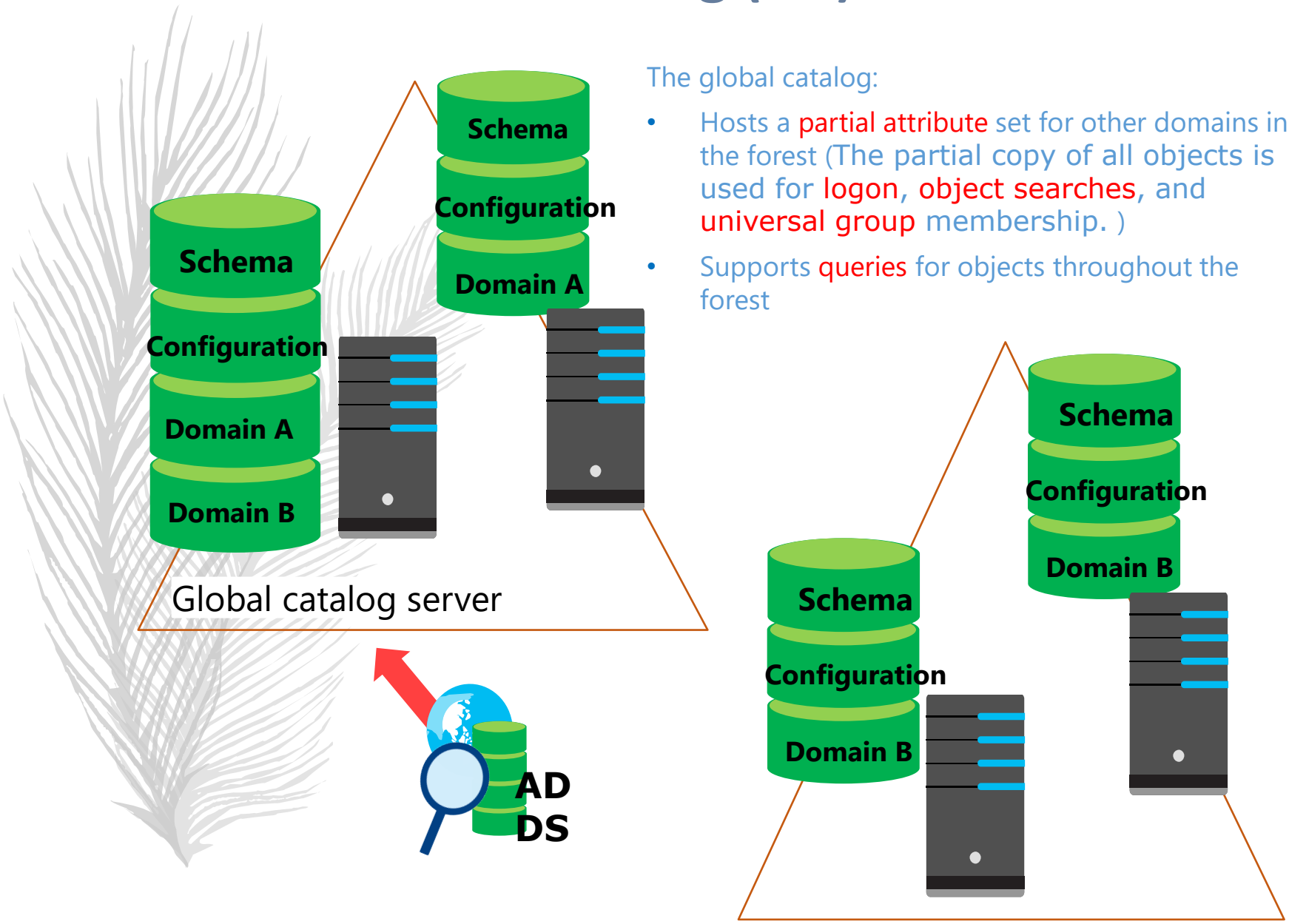
Site (cont.)

- Reasons to define a site
  - Enable a client to access network servers using the most efficient physical route
  - DC replication is most efficient when Active Directory has information about which DCs are in which locations
- One advantage of creating a site is that it sets up redundant paths between DCs
  - Paths are used for replication
- Bridgehead server
  - A DC that is designated to have the role of exchanging replication information
- Only one bridgehead server is set up per site

DCs Replicating Within and Between Sites



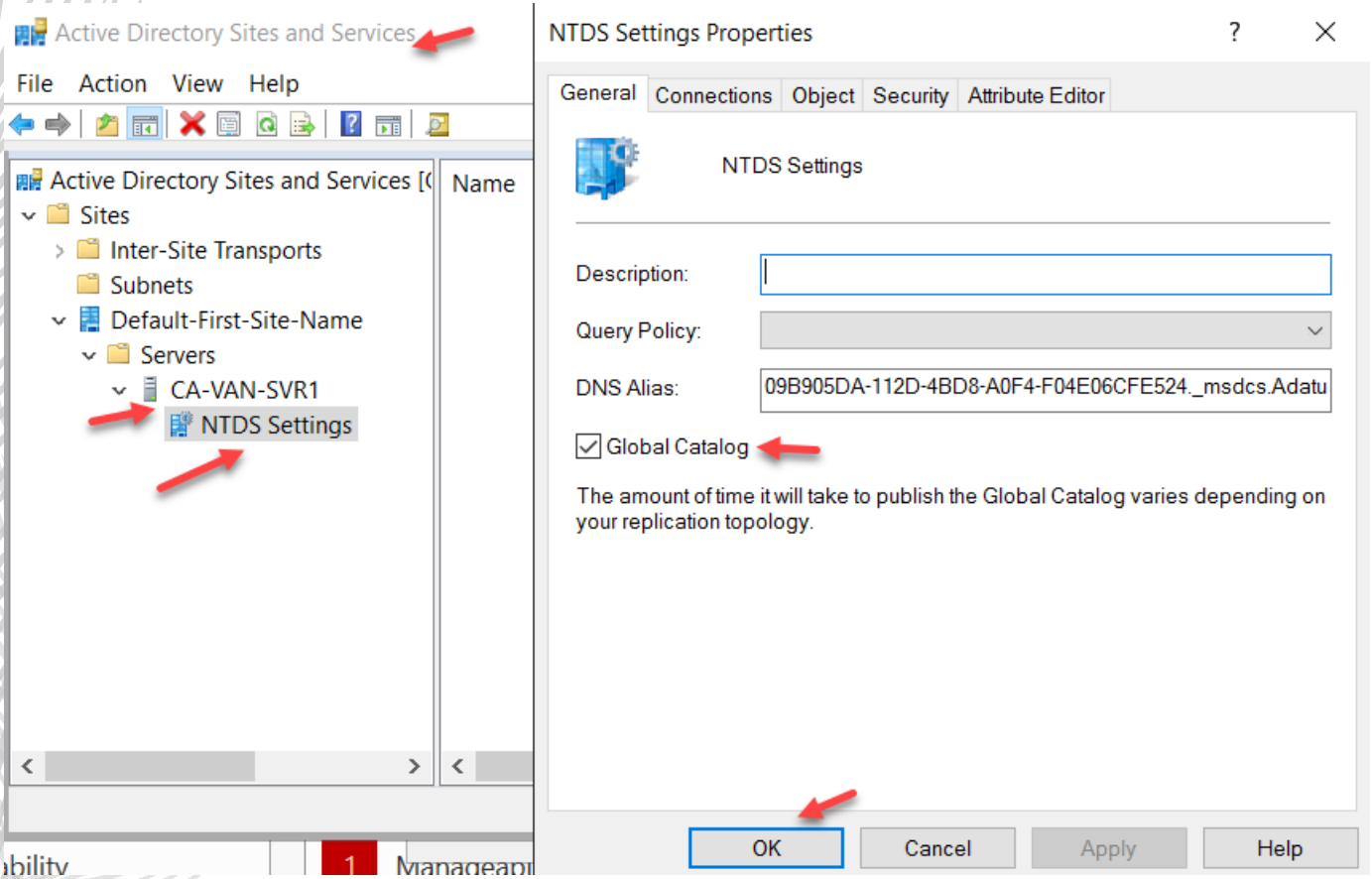
Global Catalog (GC)



The global catalog:

- Hosts a partial attribute set for other domains in the forest (The partial copy of all objects is used for logon, object searches, and universal group membership.)
- Supports queries for objects throughout the forest

Configuring the Global Catalog

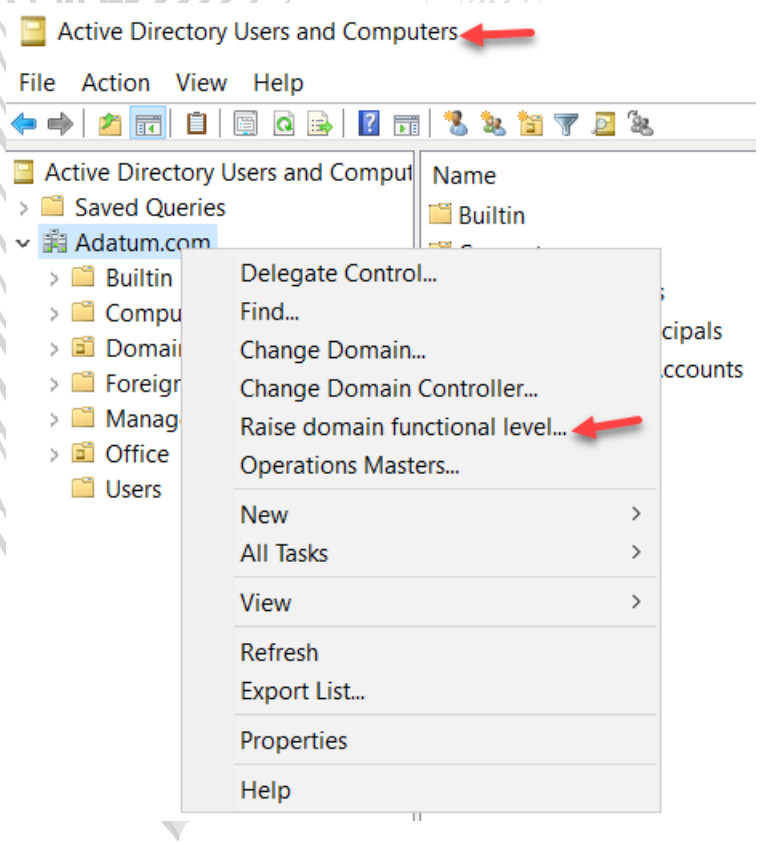


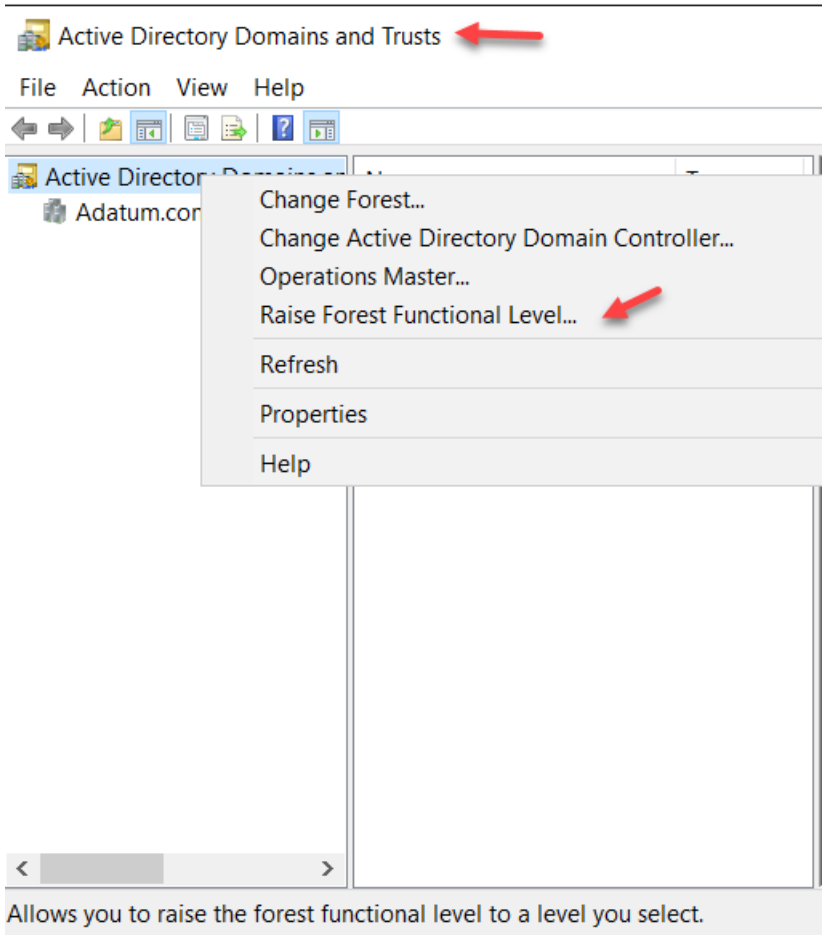
Q: Can we set all DCs as GC? Benefits vs Drawbacks

Functional Levels

- **Functional levels** are designed to provide backwards compatibility in AD DS installations with domain controllers running various versions of the Windows Server operating system.
- By selecting the functional level representing the oldest Windows version running on your domain controllers, you disable the new features so that the various domain controllers can interoperate properly.

Functional Levels (Benefits vs. Drawbacks)





Deploying AD DS

There are many variables that can affect the performance of an Active Directory installation:

- The hardware you select for your domain controllers
- The capabilities of your network
- The types of WAN links connecting your remote sites

As with most major network technologies, installing AD DS on a test network first, before you put it into actual production, is a good idea.

Installing AD DS on Server Core

- In Windows Server 2019, you can now install Active Directory Domain Services on a computer running the Server Core installation option and promote the system to a domain controller, all by using Windows PowerShell.
- For AD DS installations on Server Core, Windows PowerShell is now the preferred method.
- As with the wizard-based installation, the PowerShell procedure occurs in two phases: First, you must install the Active Directory Domain Services role; then you must promote the server to a domain controller

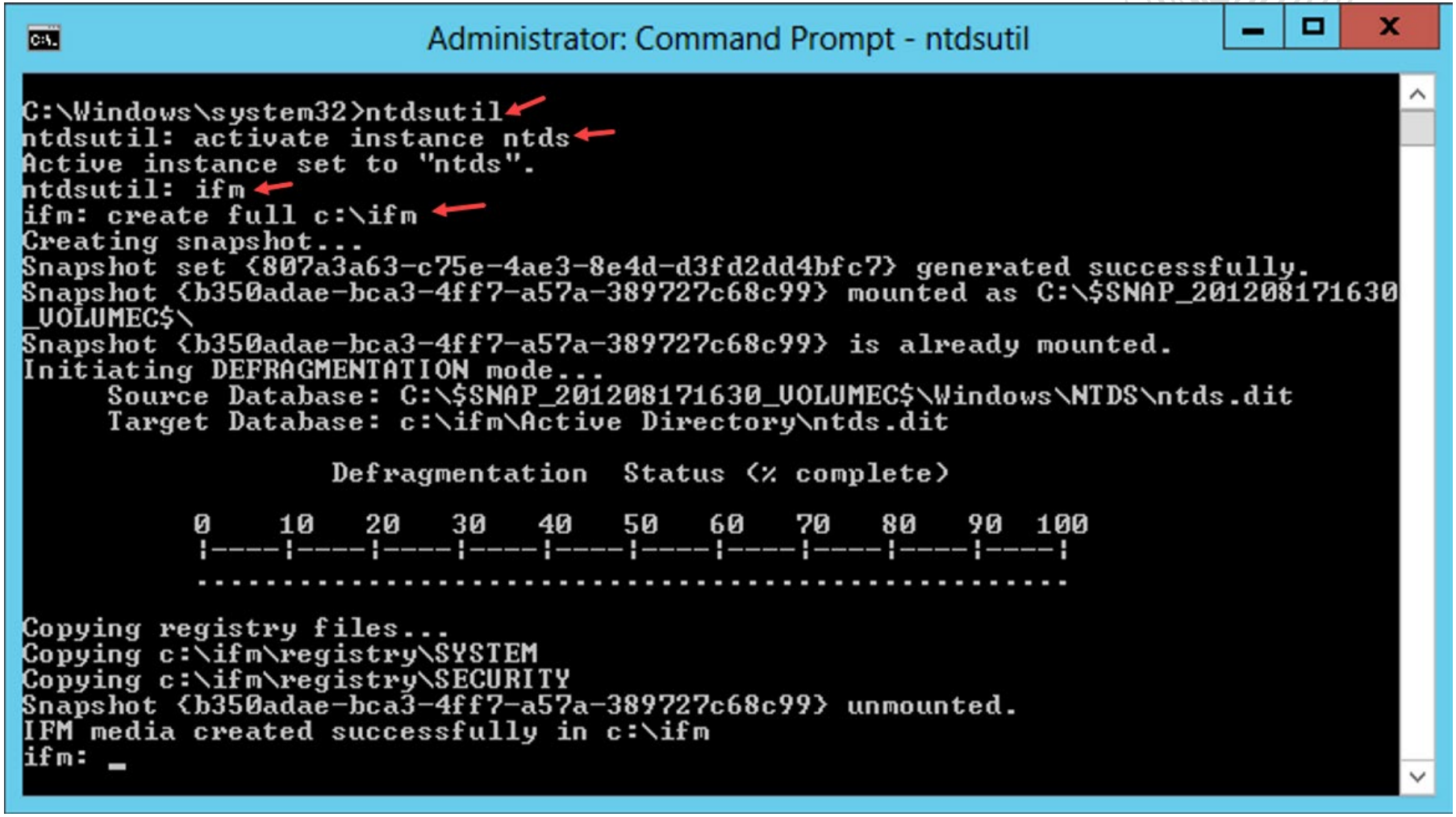
Upgrading AD Domain Services

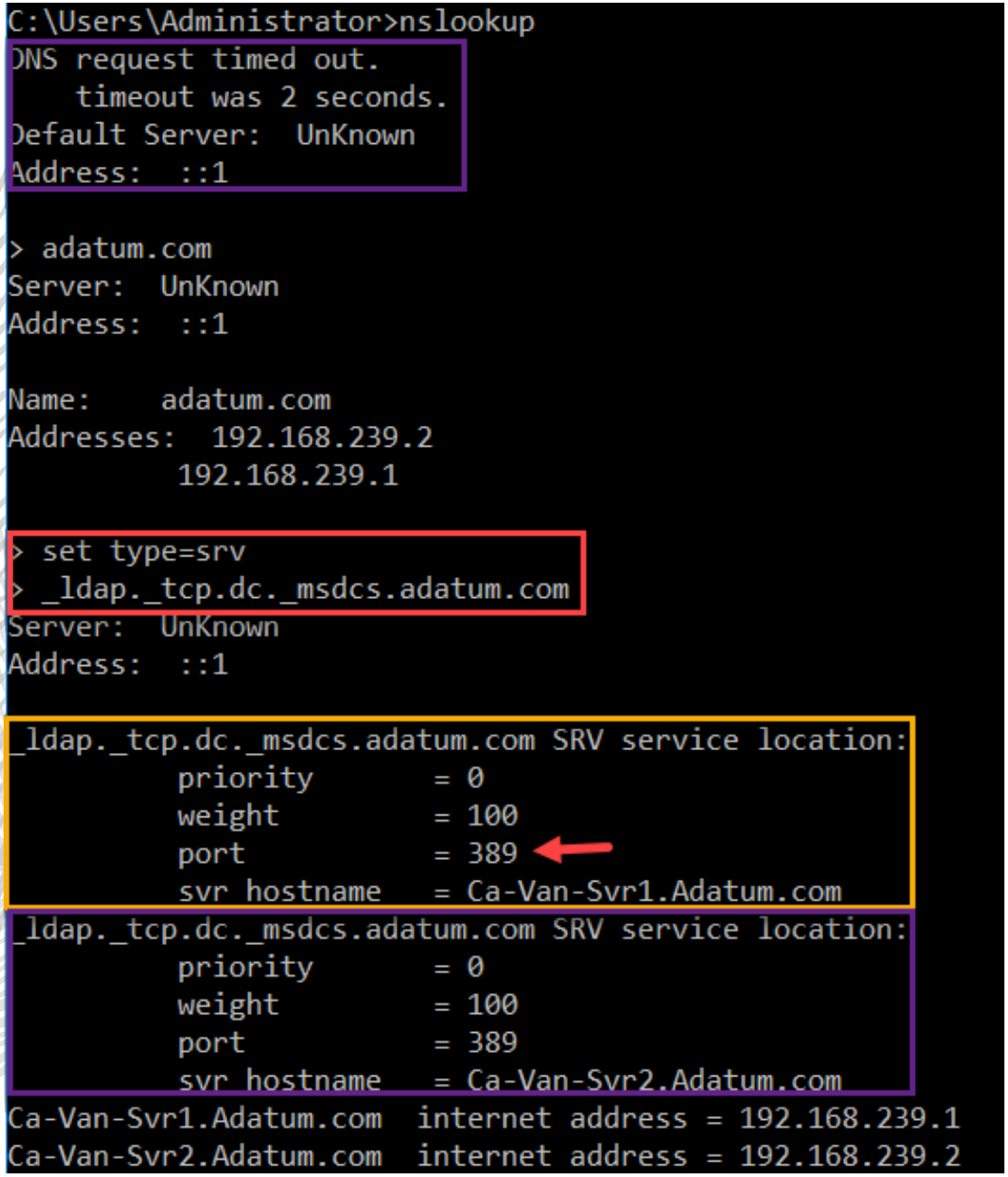
- Introducing a Windows Server 2019 domain controller onto an existing AD DS installation is quite simple.
- You can upgrade the existing down-level domain controllers to Windows Server 2019 or you can add a new Windows Server 2019 domain controller to your existing installation.
- You can upgrade a Windows Server 2008 R2 with SP1, Windows Server 2012, or Windows Server 2012 R2 domain controller to Windows Server 2019, but no earlier versions are upgradable.

Install from Media (IFM)

- **Install from media** is an option that enables administrators to streamline the process of deploying replica domain controllers to remote sites.
- Using a command line tool called **Ntdsutil.exe**, administrators can create domain controller installation media that includes a copy of the AD DS database.
- When using this installation media, the data is installed along with the database structure and no replication (**X**) is needed.

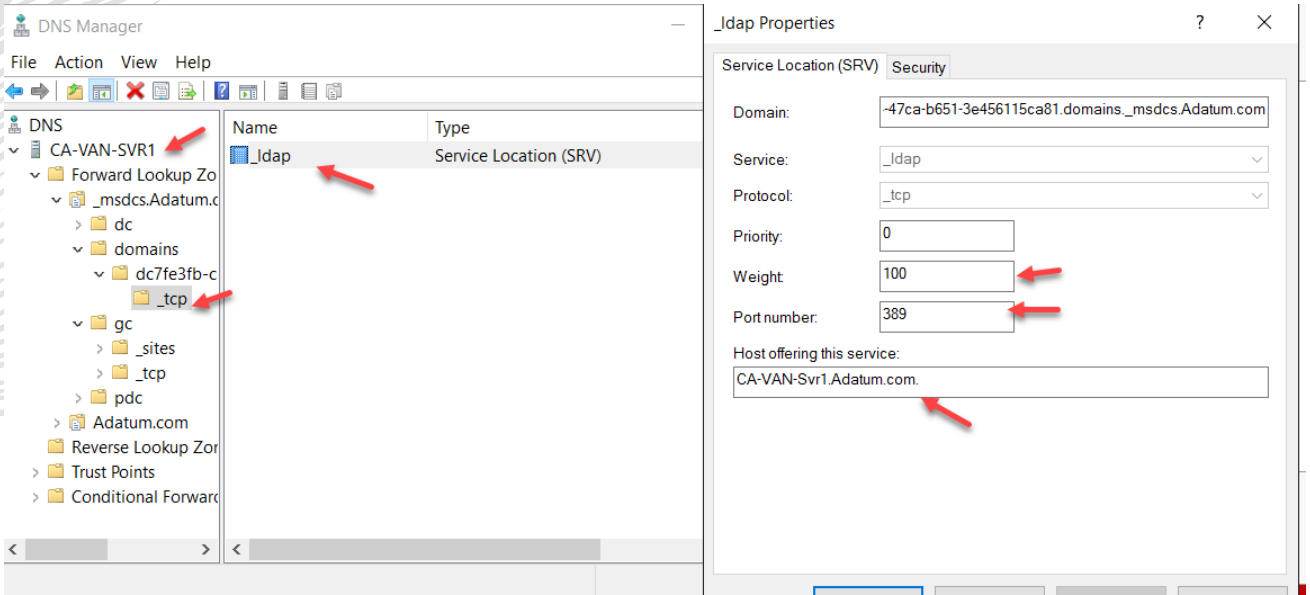
Install From Media (IFM)



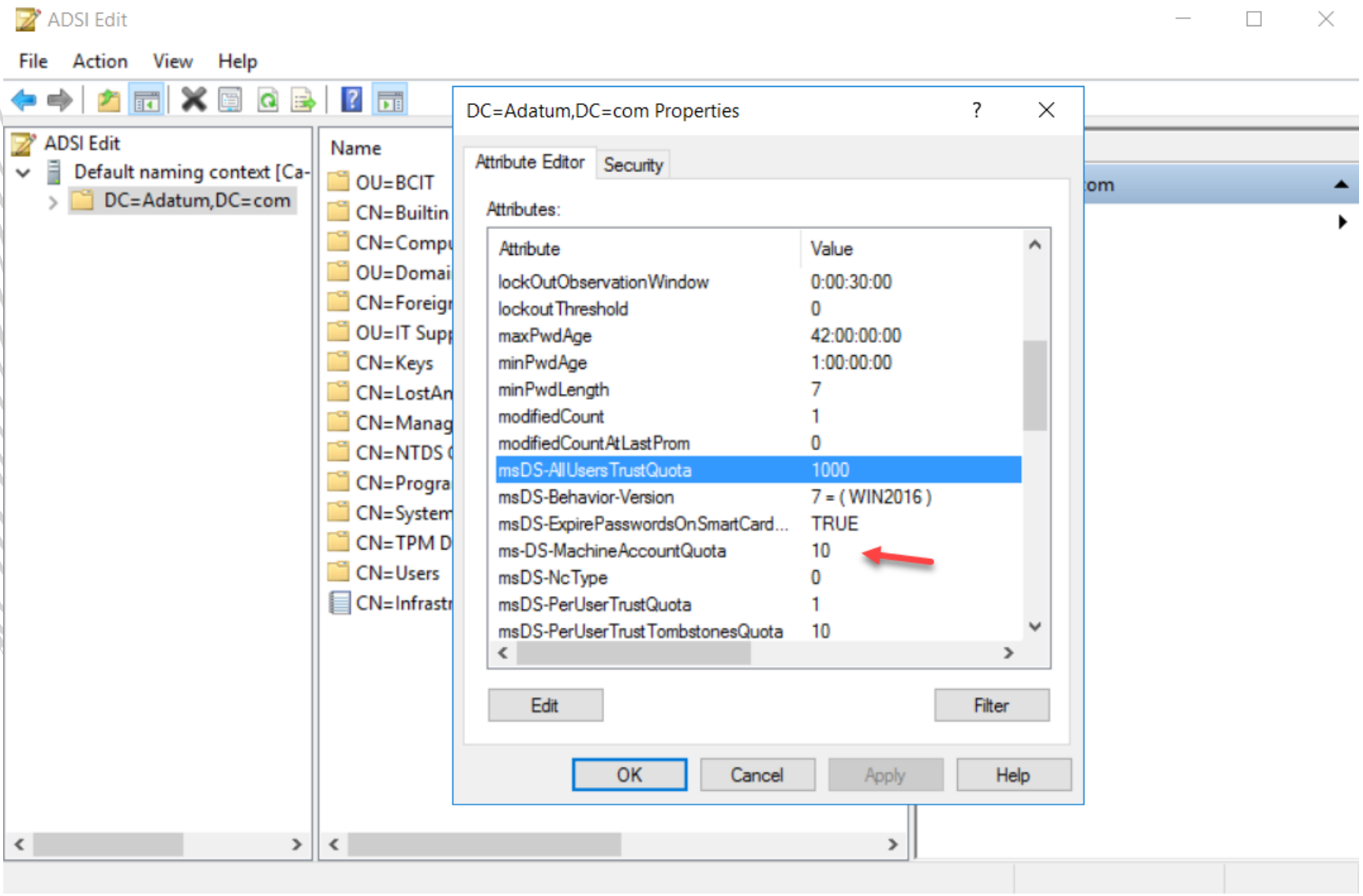

C:\Users\Administrator>nslookup
DNS request timed out.
timeout was 2 seconds.
Default Server: UnKnown
Address: ::1
> adatum.com
Server: UnKnown
Address: ::1
Name: adatum.com
Addresses: 192.168.239.2
192.168.239.1
> set type=srv
> _ldap._tcp.dc._msdcs.adatum.com
Server: UnKnown
Address: ::1
_ldap._tcp.dc._msdcs.adatum.com SRV service location:
priority = 0
weight = 100
port = 389
svr hostname = Ca-Van-Svr1.Adatum.com
_ldap._tcp.dc._msdcs.adatum.com SRV service location:
priority = 0
weight = 100
port = 389
svr hostname = Ca-Van-Svr2.Adatum.com
Ca-Van-Svr1.Adatum.com internet address = 192.168.239.1
Ca-Van-Svr2.Adatum.com internet address = 192.168.239.2

Troubleshooting DNS SRV Registration Failure

- The Domain Name System (DNS) is essential to the operation of Active Directory Domain Services.
- To accommodate directory services such as AD DS, a special DNS resource record was created that enables clients to locate domain controllers and other vital AD DS services



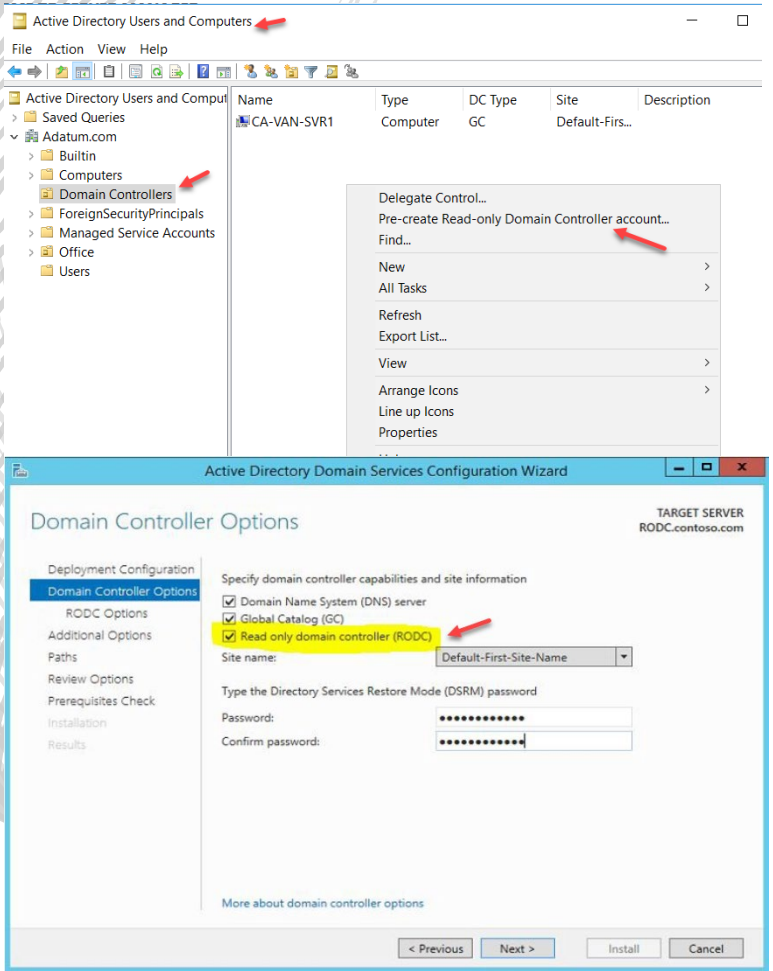
How to Restrict Domain Users to Join Computers Into Domain Windows Server

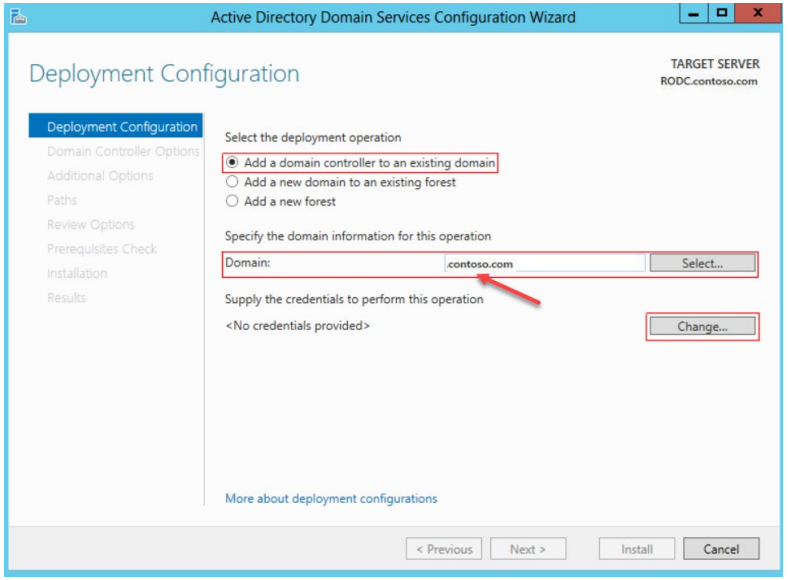


Installing and Configuring an RODC

- Windows Server 2008 introduced the ***read-only domain controller (RODC)***, which contains a full replication of the domain database.
- It was created to be used in places where a domain controller is needed but the **physical security** of the domain controller cannot be guaranteed.
- For example, it might be placed in a remote site that is not very secure and has a slower WAN link. Because it has a slow WAN link, a local domain controller would benefit the users at that site.

Installing and Configuring an RODC





Cloning a Domain Controller

- Starting with Windows Server 2012, you can safely virtualize a domain controller and rapidly deploy virtual domain controllers through cloning.
- It allows you to **quickly restore domain controllers** when a failure occurs and to rapidly provision a test environment when you need to deploy and test new features or capabilities before you apply the features or capabilities to production.

Managing Operations Masters

Operations masters, sometimes referred to as Flexible Single Master Operations (FSMO), are specialized domain controllers that perform certain tasks that can be handled only by a single domain controller in a multimaster environment.

- Primary Domain Controller (PDC) Emulator (*AD Users and Computers*)
- Infrastructure Master (*AD Users and Computers*)
- Relative Identifier (RID) Master (*AD Users and Computers*)
- Schema Master (Register: regsvr32 schmmgmt.dll)



- Domain Naming Master (*AD Domains and Trust*)

Groups

- Group objects contain users (from a single or multiple domains or OUs) who require similar access to resources or rights to perform tasks.
- Members of a group inherit rights and permissions assigned to the group.

Group Types

- Distribution groups
  - Used only with email applications
  - Not security enabled (no SID)
  - Cannot be given permissions
- Security groups
  - Security principal with a SID
  - Can be given permissions
  - Can also be email-enabled





- Security groups and distribution groups can be converted to the other type of group

Group Scopes

- **Local Groups** can contain *users, computers, global groups, domain local groups and universal groups from the same domain, domains in the same forest and other trusted domain* and can be given permissions to resources on the **local computer only**
- **Domain Local Groups** have *the same membership possibilities* but can be given permission to resources **anywhere in the domain**
- **Universal Groups** can contain *users, computers, global groups and other universal groups from the same domain or domains in the same forest* and *can be given permissions to* ***any resource in the forest (No Domain Local)***
- **Global Groups** can only contain *users, computers and other global groups from the same domain* and can be given permission to resources in the ***domain or any trusted domain*** **(No Domain Local)**

Implementing group management

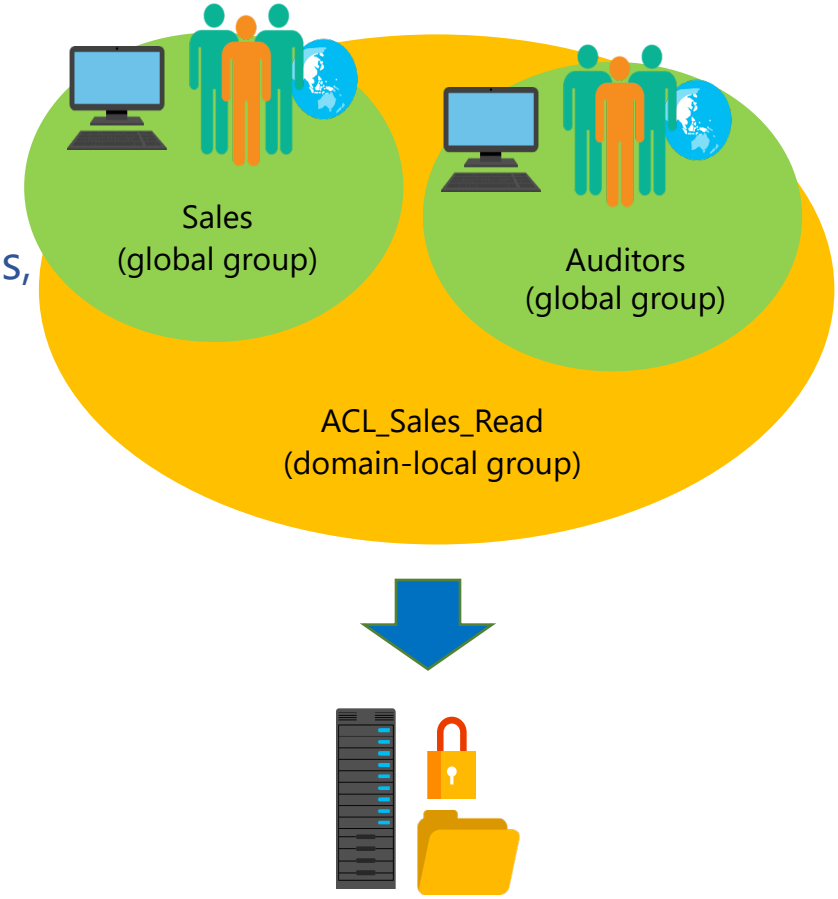
This best practice for nesting groups is known as IGDLA

I: Identities, users, or computers, which are members of

G: Global groups, which collect members based on members' roles, which are members of

DL: Domain-local groups, which provide management such as resource access which are

A: Assigned access to a resource



Questions

Adatum.com->Downtown (OU)

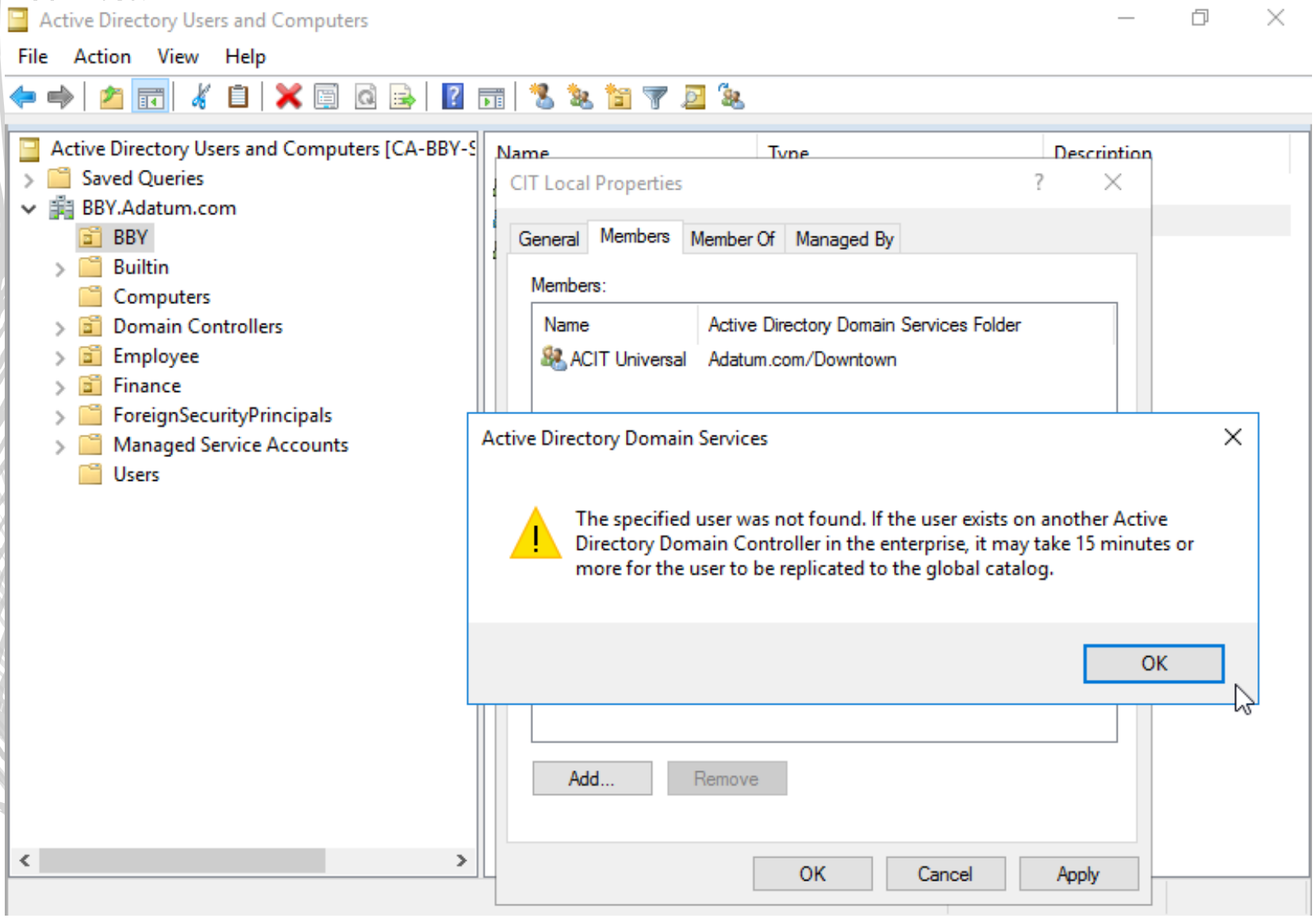
- ACIT Global (Global Group)
- ACIT Local(DL)
- ACIT Universal (Universal Group)
- ACIT Students (Universal Group)
- ACIT Faculty Global (Global Group)

Bby.adatum.com->BBY(OU)

- CIT Global (Global Group)
- CIT Local(DL)
- CIT Universal (Universal Group)

1. Can I add **ACIT Faculty Global** to **ACIT Global**?
2. Can I add **ACIT Universal** to **ACIT Global**?
3. Can I add **ACIT Local** to **ACIT Faculty Global**?
4. Can I add **ACIT Global** to **ACIT Universal**?
5. Can I add **ACIT Universal** to **ACIT Local**?
6. Can I add **CIT Global** to **ACIT Global**?
7. Can I add **CIT Local** to **ACIT Local**?
8. Can I add **ACIT Universal** to **CIT Universal**?
9. Can I add **ACIT Universal** to **CIT Local**?
10. Can I add **ACIT Global** to **CIT Universal**?

Issue


Active Directory Users and Computers
File Action View Help
Active Directory Users and Computers [CA-BBY-S
Saved Queries
BBY.Adatum.com
BBY
Builtin
Computers
Domain Controllers
Employee
Finance
ForeignSecurityPrincipals
Managed Service Accounts
Users
Name
Type
Description
CIT Local Properties
General
Members
Member Of
Managed By
Members:
Name
Active Directory Domain Services Folder
ACIT Universal
Adatum.com/Downtown
Active Directory Domain Services
The specified user was not found. If the user exists on another Active Directory Domain Controller in the enterprise, it may take 15 minutes or more for the user to be replicated to the global catalog.
OK
Add...
Remove
OK
Cancel
Apply

Replication

```
C:\Users\Administrator>repadmin /replicate ca-van-svr1 ca-van-svr2 dc=adatum,dc=com
Sync from ca-van-svr2 to ca-van-svr1 completed successfully.


C:\Users\Administrator>
```

Active Directory (Summary)

- The *physical* components that make up Active Directory include the following:
- ***Domain controllers***:
  - Are servers that host the AD DS database (Ntds.dit) and SYSVOL
  - Host the Kerberos authentication service and KDC services to perform authentication
  - Have best practices for:
    - *Availability:*
      - *Use at least two domain controllers in a domain*
    - *Security:*
      - *Use an RODC or BitLocker*
- ***Site***: A group of IP subnets that are connected at high speed.